

SEMESTER – I

PAPER – I

PSYCHOLOGY AND SOCIOLOGY

**Instruction Hrs:
Theory – 150 hrs**

Section – A: Psychology (90 hrs)

I. General Psychology (50 hrs)

1. Definition of psychology
 - a. Science of mind, consciousness and behavior
 - b. Scope and branches of psychology
2. Methods of introspection, observation and experimentation
3. Hereditary and environment
 - a. Relative importance of hereditary and environment
 - b. Physical characteristics intelligence and personality
 - c. Nature Vs. nurture controversy
4. Learning: Types of learning
 - a. Trial and error
 - b. Classical learning
 - c. Instrumental learning
 - d. Insight for learning
5. Memory
 - a. Steps of memory
 - b. Measurement of memory
 - c. Causes of forgetting
 - d. Concepts of STM & LTM
6. Perceptual process
 - a. Nature of perceptual process
 - b. Structural and functional factors in perception
 - c. Illusion and hallucination
7. Emotion
 - a. Emotion and feeling
 - b. Physiological changes
 - c. Theories of emotion (James – Lange and Cannon – Bird)

8. Motivation
 - a. Motive, need and drive
 - b. Types of motive: Physiological, psychological and social
9. Intelligence
 - a. Definitions: theory and assessment
10. Personality: Definition, Types and measurements

II. Child Psychology (10 hrs.)

1. Concept of child psychology
 - a. Meaning: nature, and subject matter of child psychology
 - b. Practical importance of studying child psychology for rehabilitation professionals
2. Methods of studying child development
 - a. Baby biography
 - b. Case history
 - c. Behavior rating

III. Industrial Psychology (30 hrs.)

1. Human Engineering
 - a. Importance of human engineering
 - b. Development of human engineering
 - c. Problems in human engineering
2. Decision making
 - a. Process and steps in decision making
 - b. Individual decision making
 - c. Decision making in organization
3. Stress and mental health
 - a. Causes and reaction to stress
 - b. Stress management
4. Work culture, moral and rewards of work discipline
5. Guidance and counseling
 - a. Meaning, types and objectives of counselor

Section – B: Sociology (60 hrs)

1. Introduction: Definition of sociology, sociology as a science, uses of the study of sociology, application of knowledge of sociology in physiotherapy and occupational therapy.
2. Sociology and health: Social factors affecting health status, Social consciousness and perception of illness, social consciousness and meaning of illness, decision making in taking treatment, institutions of health their role in the improvement of health their role in the improvement of health and the people.
3. Socialization: Meaning of socialization, influence of social factors on personality, socialization in hospital and socialization in rehabilitation of patients.
4. Social groups: Concepts of social groups influence of formal and informal groups on health and sickness, the role of primary groups in the hospital and secondary groups in the hospital and rehabilitation setting.
5. Family: Influence of family on human personality, discussion of changes in the functions of a family, influence of family on the individual's family and psychosomatic disease.
6. Culture: Components of culture, impact of culture of human behaviors, cultural meaning of sickness, response to sickness and choice of treatment (role of culture as social consciousness in moulding the perception of reality). Culture induced symptoms and diseases, sub-culture of medical workers.
7. Caste System: Features of the modern caste system and its trends.
8. Social: Meaning of social control, role of norms, folkways, customs morals religion, law and other means of social control in the regulation of human behavior, social deviance and disease.
9. Social problems of the disabled: Consequences of the following social problems in relation to sickness, disability, and remedies to prevent these problems
 - a. Population explosion
 - b. Beggary
 - c. Juvenile delinquency
 - d. Prostitution
 - e. Alcoholism
 - f. Problems of women in employment
10. Social security: Social security and social legislation in relation to the disabled
11. Social worker: The role of a medical social worker

PAPER – II
THERAPEUTICS – I

Instruction Hrs:
Theory – 150 hrs
Practical – 100 hrs

Section – A: Exercise Therapy I (125 hrs)

Theory (75 hrs)

Practical (50 hrs)

1. Basic Physics in Exercise Therapy: Mechanics, Force, Gravity, line of gravity, centre of gravity in human body, Base, equilibrium, Axes and planes, Mechanical principles - Lever, examples in human body, pendulum, spring, friction, weight & pulley circuit.
2. Massage: Definition of massage, types of massage, general effects and uses of massage, local effects of individual manipulation (Physiological effects), contraindications, techniques of application of all manipulations – stroking, Effleurage, Kneading and picking up, skin rolling (back), clapping tapping, friction, etc.

Practical:

Demonstration and practice of all types of massage manipulation: Stroking, Effleurage, Kneading - circular kneading, thumb kneading, finger kneading (to joints) etc. Picking up, skin rolling (back) clapping etc.

The above various types of manipulation should be demonstrated and practiced to upper limbs, lower limbs back and face appropriately.

3. Introduction to Exercise therapy
4. Starting Positions: Fundamental starting positions - standing, sitting, kneeling, lying, and hanging. All the derived positions of the above five fundamental starting positions. Muscle work for all the fundamental starting positions. Derived positions
5. Classifications of movements in details

Active Movement: Voluntary movements, involuntary movements, Passive movements
6. Voluntary movements: Free exercise, assisted exercises, resisted exercise
7. Assisted exercises: Technique and uses.
8. Free exercises: Classification, technique, effects of free exercises on various systems etc.

9. Resisted exercise: Technique and types of resistance, SET system (Heavy resisted exercises, Oxford method, Delorme's method, method, McQueen's method)
10. Relaxed passive movements: Basic knowledge of classification of relaxed passive movements, definition technique, effects and uses of relaxed passive movements.

Practical:

Demonstration and practice of relaxed passive movements to upper limb, lower limb and spine.

11. Bed rest: Complications, effects of physiological standing & use of tilt table.
12. Suspension therapy: Principles of suspensions, types of suspension therapy, effects and uses of suspension therapy - their application either to mobilize a joint to increase joint range of motion or to increase muscle power explaining the full details of components uses for suspension therapy.

Practical:

Demonstration and practice of putting suspension to shoulder joint, elbow joint in upper limb, hip joint and knee joint in lower limb for all movements.

13. Posture: Types, factors responsible for good posture and poor posture, principles of development of good posture, faulty posture, correcting technique.

Section – B: Electrotherapy (125 hrs)

Theory (75 hrs)

1. Electrical fundamentals
 - a. Physical principles - Structure and properties of matter, molecular atom, proton, neutron, electron, ion, etc.
 - b. Electrical Energy: Nature of electricity - Current - Static electricity current - Electric potentials generated by cell - Gem's Law, Joule's Law.
 - c. Magnetic Energy: nature and property of a magnet, magnetic induction, Snow rule - Maxwell's cork & screw rule.
 - d. Electromagnetic induction - Principle and working of choke coil - Transformer - Rectification of AC to DC. Metal Oxide rectifier, Semiconductor - Diode and Triode.
 - e. Valves - Principles of working - condenser - principle - Details of charging and discharging etc. Transistors measurement of current intensity, EME and power - moving coil mill ammeter and voltmeter.
 - f. Wiring of components in series and parallel. Distribution of electrical energy – Earth shock and electrical shock, safety devices.

2. Low frequency currents
 - a. Nature and principles of production of muscle stimulating currents used for treatment, Therapeutic electric stimulation, Iontophoresis, High voltage galvanic current, Rectifying currents.
 - b. Electrogenic membrane response - chemo responsive electrogenic system, Neuromuscular junction - synapses - Muscle electrogenic - Electro physiology of C.N.S.
 - c. Constant direct and interrupted direct currents - Modified current - Physiological and therapeutic effects and uses, Technique and method of application - precautions against dangers.
 - d. Accident and treatment of them if they occur.
 - e. Muscle stimulating current - To innervated and denervated muscle - Long and short duration various pulses Accommodation.
 - f. Principles of electro diagnosis - Strength duration curve - Chronaxie and Rheobase - Their relationship etc.
3. Therapeutic heat: Definition, production, physiological & therapeutic effect, uses contra indication, technique of application of following.
 - a. Moist heat
 - b. Paraffin wax bath
 - c. Contrast bath
 - d. Whirlpool bath
 - e. Fluido therapy
 - f. Electric heating pads

Practical (50 hrs)

Low frequency current treatment:

- Preparation of electro-therapy, preparation of apparatus, patient treatment technique. Following treatment techniques should be demonstrated and practice by students:
 - Stimulation of motor points
 - Stimulating the muscles of extremity, back and face through the motor points
 - Quadriceps inhibition
 - Deltoid inhibition
 - Faradism under pressure
 - Faradism under tension
 - Nerve conduction method
 - Diagnostic tests:
 - FG Test
 - SD Curve

- Fatigue Test

- Uses of surged faradism and interrupted Galvanism in various peripheral nerve lesions:
 - Neuropraxia
 - Axonotemesis
 - Neurotemesis
- Pain-physiology, pain modulation & Tens

SEMESTER – II

PAPER – I

ANATOMY

Instruction Hrs:

Theory – 100 hrs

Practical & Demonstration – 100 hrs

General Anatomy (16 hrs)

Introduction to anatomy:

Cell : Parts, name of cytoplasmic organelles and inclusion with their functions

Epithelium : Types with examples and light microscopic structure.

Connective tissue : Classification with emphasis to tendon.

Cartilage : Types with example.

Bone : Types with examples, types of ossification (Stages of ossification not required), blood supply, fracture repair.

Joints : Classification with example, emphasis to synovial joint.

Muscles : Types (details of E.M picture not required)

Nervous tissue: Structure of a neuron, synapse reflex arc, degeneration and regeneration of nerve.

REGIONAL ANATOMY

Superior extremity (32 hrs)

Theory (14 hrs)

Axilla, brachial plexus, shoulder joint, sternoclavicular joint, axillary lymph nodes, elbow joint, superior radio-ulnar joint, nerves of arm and fore arm, synovial bursa of hand and palmar space, ulnar nerve in hand, cutaneous distribution according to dermatome, clinical anatomy, surface anatomy.

Practical/demonstration (18 hrs)

Pectoral Region, Axilla, Scapula and Clavicle, Humerus, Muscles of arm (Front & Back) Radius, Front of forearm, ulna, back of forearm, muscle of palm a arterial arches, articulated hand (carpals and meta carpals name and arrangements in order only).

Inferior extremity (32 hrs)**Theory (13 hrs)**

Lumbar plexus, inguinal group of lymph nodes, hip joint, femoral triangle and femoral sheath, knee joint, venous drainage of inferiors extremity, sciatic nerve and its distribution, obturator nerve, arches of foot, mid tarsal and sub talar joint, cutaneous distribution according to myotome, clinical anatomy, surface markings.

Practical/demonstration (19 hrs)

Hip bone, Gluteal Muscles, Femur, front of thigh, back of thigh, medial side of thigh, tibia, anterior compartment of leg, fibula, lateral compartment of leg, back of leg, articulated foot(Identification of tarsal meta tarsal only).

Abdomen and Pelvis (34 hrs)**Theory (15 hrs)**

Abdominal wall, inguinal canal, stomach, liver, pancreas, kidney with ureter and spleen, small intestine, large intestine, abdominal aorta, portal vein, diaphragm, sacral plexus, sacro-iliac joint, intervertebral disc.

Practical/demonstration (19 hrs)

Abdominal viscera, sacrum, bony pelvis, viscera of pelvis and blood vessels.

Thorax (16 hrs)**Theory (07 hrs)**

Thoracic cage and mediastinum, heart with its internal and external features, blood vessels, typical spinal nerve, typical intercostals space, mechanism of respiration, surface markings of heart and lungs.

Practical/demonstration (09 hrs)

Superior mediastinal structures, sternum, ribs (only general features), vertebrae (Identification, general features, functional components, development, vertebral column with weight transmission), heart, pleura & lungs.

Head & Neck (22 hrs)

Theory (11 hrs)

Temporo mandibular joint, atlanto-occipital and atlanto-axial joint, cutaneous distribution of trigeminal nerve.

Practical/demonstration (11 hrs)

Mouth cavity, nasal cavity, pharynx and larynx (Parts, sensory distribution), cranial bones (Identification of individual bone general features, different foramina in relation to cranial nerve, cranial fossae and their relations to brain and hypophysis).

Identifications of anterior and posterior triangles of neck with their contents.

Nervous system (24 hrs)

Theory (12 hrs)

General introduction and classification, automatic nervous system (Idea about sympathetic and para sympathetic with their difference in distribution and function).

Spinal cord with its meninges, spinal Reflex, Pyramidal and extrapyramidal tracts (detail Nucleus not required) Blood supply.

Parts of brain, meninges, Gross discussion of Hind Brain, Mid Brain (cranial nerve nucleus position should be mentioned).

Fore brain - cerebral hemisphere, functional areas and blood supply.

Practical/demonstration (12 hrs)

Spinal cord and parts of brain

Cranial nerve (12 hrs)

Names in order, individual cranial nerve distribution, idea about upper motor neuron and lower motor neuron, applied anatomy.

Histology Practical (12 hrs)

Epithelium (Simple, Compound)

Connective tissue (Cartilage & Bone)

Muscle (smooth & skeletal)

Nervous tissue (Neuron)

Blood vessels (Large artery and vein)

PAPER – II
PHYSIOLOGY

Instruction Hrs:
Theory – 50 hrs
Practical & Demonstration – 50 hrs

General Physiology

1. Introduction and scope of physiology.
2. Cell and tissue - its structure, principal constituents, properties and functions including cell division.
3. Body fluid
 - a. Blood: composition and general function of plasma. Blood cells - structure and function - red blood cells, white blood cells - including numbers and approximate length of life - position, structure and function of cells of reticulo endothelial system.
 - b. Blood clotting including bleeding time and clotting time, factors accelerating or slowing the process. Blood groups and their significance, Rh factor, Hemoglobin and E.S.R
 - c. Formation of blood, tissue fluid and lymph.
4. Cardio- Vascular System
 - a. Structure and properties of Heart Muscles and nerve supply of Heart Structure and function of arteries, capillaries and veins.
 - b. Cardiac cycle and Heart sound.
 - c. Cardiac output measurements, factors affecting Heart Rate and its regulation, cardiovascular reflexes.
 - d. Blood pressure, its regulation, physiological variation, peripheral resistance, factors controlling Blood pressure.
 - e. Hemorrhage.
5. Respiratory System
 - a. Mechanism of Respiration, changes in diameters of thorax intra-pleural and intrapulmonary pressure.
 - b. Quantities of lung volume tidal and residual volume, vital capacity.
 - c. Gaseous interchanges in lung and tissues.
 - d. Control of respiration - nervous and chemical significance of changes in rate and depth, transportation of oxygen and carbon dioxide.
 - e. Respiratory states - anoxia, asphyxia, cyanosis, acclimatization

6. Digestive System

- a. General arrangement of alimentary canal, liver, pancreas, position structure and functions.
- b. Nutrition and Diet - carbohydrates, protein, fat, salts, water, vitamins and minerals digestion, absorption and metabolism.

7. Reproductive system

- a. Sex determination and development of puberty, male sex hormones, spermatogenesis, female sex hormones, menstrual cycle, ovulation, pregnancy, function of placenta, lactation.

8. Excretory System

- a. Gross and minute structure of kidney, renal circulation mechanism of formation, of urine, glomerular filtration rate and tubular function, renal function and renal tests. Physiology of micturation.

9. Endocrine system

- a. Structure and function of pituitary (anterior and posterior). Thyroid, parathyroid, adrenal cortex, adrenal medulla, Thymus and pancreas. Blood sugar regulation.

10. Skin - structure and function

11. Neuromuscular physiology

- a. Cell membrane - ionic and potential gradient and transport Muscle - types of muscular tissue - gross and microscopic structure - function. Basis of muscle contraction - changes in muscle contraction, electrical - biphasic and monophasic action potentials, chemical, thermal and physical changes, isometric and isotonic contraction.
- b. Motor units and its properties - clonus, tetanus, all or none law, Fatigue.
- c. Nerve - Gross and microscopic structure of nervous tissue, one neuron Generation of action potential - Nerve impulse - conduction. Neuromuscular Junction.
- d. Degeneration - Regeneration of peripheral nerves Wallerian degeneration, Electrotonus and Duffus law. Types and properties of receptors, types of sensations, synapse, reflex and its properties occlusion, summation, subliminal fatigue etc.
- e. Tracts - Ascending and descending and extrapyramidal tracts.
- f. Functions of E.E.G
- g. Functions of cerebral cortex, cerebrum, cerebellum, Basal ganglia. Thalamus connection and functions.
- h. Reticular formation - tone, posture & equilibrium, autonomic nervous system.
- i. Special senses eye - errors of refraction, lesions of visual pathways. Speech and its disorders. Ear and vestibular apparatus, taste, olfaction somatic sensations.

12. Work physiology

- a. Neuromuscular activity, human movement, physiological mechanism in movement behavior, skill, strength, endurance, analysis of movement.
- b. Circulatory and respiratory response to exercises and work the heart blood circulation, body fluid changes, pulmonary ventilation, gas exchange and transport.
- c. Effects of exercise and work on other body functions.

PAPER – III
BIOCHEMISTRY

Instruction Hrs:
Theory – 50 hrs

1. Biophysics
Concepts of pH and buffers, Acid - base equilibrium, osmotic pressure and its physiological applications.
2. Cell
Morphology, Structure and functions of cell, cell membrane, Nucleus, Chromatin, Mitochondria, Endoplasmic reticulum, Ribosome.
3. Carbohydrates
Definition, Functions, Sources, Classification, Monosaccharides, Disaccharides, Polysaccharides, Mucopolysaccharides and its importance.
4. Lipids
Definition, functions, sources, classification, simple lipids, compound lipids, derived lipids. Saturated and unsaturated fatty acids. Essential fatty acids & their importance, Blood lipids and their implications, cholesterol and its importance.
5. Proteins
Definition, Sources, Function, Classification, Simple protein, Conjugated proteins and Derived proteins, Properties and reaction of proteins.
6. Nucleic acids
Structure and functions of DNA, RNA, Nucleotides, Nucleotides Genetics code, Biologically important Nucleotides.
7. Enzymes
Definition, Classification, Mode of action, factors affecting enzyme action, Clinical importance of enzymes.
8. Vitamins
Classifications, Fat soluble vitamins A, D, E, K, Water soluble, Vitamins-B Complex and Vitamin C. Daily requirement, Physiological functions and diseases of vitamin deficiency.
9. Bioenergetics
Concept of free energy change, Exogenic reaction and endogenic reactions, Concepts regarding energy rich compounds, Respiratory chain and Biological oxidation.

10. Carbohydrate Metabolism

Glycolysis, HMP shunt pathway, TCA Cycle, Glycogenesis, Glycogenolysis, Gluconeogenesis, Maintenance of blood Glucose, interconversion of different sugars.

11. Lipid Metabolism

Fatty acid oxidation, Fatty acid synthesis, Metabolism of cholesterol, Ketone bodies, Atherosclerosis and obesity.

12. Protein Metabolism

Transamination, Transmethylation, Deamination, Fate of ammonia, Urea synthesis and synthesis of creatinine, Inborn Errors of Metabolism.

13. Water and Electrolytes

Fluid compartments, Daily intake and output, Dehydration, sodium and Potassium metabolism.

14. Mineral Metabolism

Iron, Calcium, Phosphorus, Trace elements.

15. Nutrients

Nutritional aspects of carbohydrate, fat and proteins. Balance diet, metabolism in exercise and injury, diet for chronically ill and terminally ill patients.

16. Connective Tissue

Mucopolysaccharides, connective tissue, proteins, glycol proteins, chemistry and metabolism of bone and teeth, metabolism of skin.

17. Nerve Tissue

Composition, metabolism, chemical mediator of nerve activities.

18. Muscle Tissue

Structure, metabolism of muscle, muscle contraction.

19. Hormone

General characteristics and mechanism of hormone action, insulin, glucagon, thyroid and parathyroid hormone, cortical and sex hormone.

20. Isotopes

Isotopes and their role in diagnosis and treatment of disease.

PAPER – IV
THERAPEUTICS – II

Instruction Hrs:
Theory – 150 hrs
Practical – 100 hrs

Section – A: Exercise Therapy II (125 hrs)

Theory (75 hrs)

Practical (50 hrs)

1. Muscle Strength

Anatomy and physiology of muscle tissue, causes of muscle weakness/paralysis, prevention of muscle weakness/paralysis. Types of muscle work and contractions, Ranges of muscle work. Muscle assessment M.R.C. grading Principles of muscle strengthening/Re - education, Early Re - education of paralyzed muscles.

Practical:

Demonstration and practice of strengthening/re - educating weak/paralyzed muscle of both upper and lower extremity muscles, (individual/group muscles) abdominal muscle exercises, spinal extension exercise, free exercises.

2. Relaxation

Technique of relaxation, principles obtaining relaxation in various position, effects & uses.

3. P.N.F basic theory of proprioceptive

Neuro muscular facilitation techniques. Different types of PNF techniques, their effects and uses.

4. Hydrotherapy

Introduction, various types of hydrotherapy units construction and equipments used in hydrotherapy. Principles, indications, contraindications, effects and uses in hydrotherapy. Precaution towards patients, towards therapist, equipment unit etc.

5. Traction

Manual and mechanical traction. Biomechanics of traction. Biomechanics of traction. Physiological effects and therapeutic uses. Technique of application.

6. Joint movement

Classification of joint movements causes for restriction of joint movement, prevention of restriction of joint range of motion etc. Principles of mobilization of a joint in increasing its range of motion. Technique of mobilization of stiff joints. Goniometry.

Practical:

Demonstration and practice of techniques to improve joint range of motion, of hip joint, knee joint, ankle foot n lower limb, shoulder joint, elbow joint, radio - ulnar joint, wrist in upper limb.

Demonstration and practice of free exercises to improve joint range of motion (small joints, e.g. hand, finger, toes etc).

Passive stretching: technique of passive stretching to sternocleidomastoid muscles, shoulder abductors, flexors, elbow flexors, supinator, wrist and finger flexors in upper limb. Passive stretching to hip flexors, abductors, Iliotibial band, tensor-fascia-lata, Quadriceps, knee flexors, tendo achilis etc. in lower limb.

7. Co-ordination exercises

Definition of co-ordinated movements in co-ordinated movements. Factors of co-ordinated movements, causes of in-coordination, principles of re-education of co-ordinated movements, technique of co-ordination exercise.

Practical:

Demonstration and practice of technique of Dr. Frenkel's exercise to improve co-ordination.

8. Gait

Analysis of normal gait with muscle work, various pathological gaits.

Practical:

Demonstration of various pathological gaits. Demonstration of various pathological gaits.

9. Crutch gait

Introduction, crutch measurement, crutch balance, various types of crutch gait in details. Measurement of crutches, walking aids, strengthening of crutch muscles, crutch balance, demonstration and practice to all crutch gaits.

10. Breathing exercises

Physiology of respiration, types of exercises, techniques of various types of breathing exercise, its effects and uses.

Practical:

Demonstration and practice of Diaphragmatic breathing localized expansion exercises.

11. Individual, group and mass exercise. Maintenance exercise plan of treatment, tables and schemes.

Books References (Both theory and practical)

1. Principles of exercise therapy – by Dena Gardiner
2. Progressive exercise therapy – by Coson & Collision
3. Human movement – by Callie
4. Exercises I water – by Doffield
5. Practical exercise therapy – by Hollis. M
6. Muscles testing – by Daniels.
7. Principles & practices of therapeutic massage – by A.G. Sinha
8. Muscle testing – by Kendal.

Section – B: Exercise Therapy II (125 hrs)**Theory (75 hrs)****Practical (50 hrs)**

1. Physics of high frequency currents - production of high frequency currents principles - Bio Physics of heat, physiology of heat and cold. Physiological and therapeutic effects and uses, technique of treatment, dangers and precautions contraindication etc. of the following:
 - a. Short wave diathermy
 - b. Ultrasound
 - c. Micro wave diathermy
2. Medium frequency current: Definition, production (Brief), physiological and therapeutic effects, uses, contra indication, technique of application of followings:
 - a. Interferential current
 - b. Russian current
3. Actinotherapy: Definition, production (Brief), physiological and therapeutic effects uses, contra indication, technique of application of following:
 - a. Infrared radiation
 - b. Laser
 - c. Ultraviolet radiation
 - d. Helio therapy
4. Cryotherapy: Principles, physiological effects, uses, techniques of application and contra indication of followings:
 - a. Cold packs
 - b. Ice massage
 - c. Commercial cold packs
 - d. Ice towels
 - e. Cold compression units
 - f. Evaporating sprays

5. Electromyography & bio feedback: Basic principles of amplifiers, Oscillators, cathode ray tube, Records, Sigma processing, display devices and indication their principles and uses in Electromyography.
6. Principle and application of Biofeedback & functional electrical stimulation.

Practical:

1. Short wave diathermy – setting up of apparatus including selection of method and electrodes. Technique – preparation of patient – checking contra indications – Application of SWD for various conditions and various parts of the body – These must be practiced by the students.
2. Microwave diathermy – Same as above
3. Ultrasonic: Setting up of apparatus – selection of dose – Technique of application of various conditions and to various parts of the body.

Books References (Both theory and practical)

1. Clayton's electrotherapy and actinotherapy
2. Principles and practice of electrotherapy – by Kahn
3. Electrotherapy of Wolf
4. Electro therapy explained – John low and Annreed

PAPER – V

ENVIRONMENTAL STUDIES

Instruction Hrs:

Theory – 45 hrs

Field work – 5 hrs

1. The multidisciplinary nature of environmental studies
 - a. Definition, scope and importance
2. Natural Resources
 - a. Renewable and non-renewable sources. Natural resources and associated problems
 - b. Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forests and tribal people.
 - c. Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dam's benefits and problems
 - d. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies.
 - e. Food resources: World food problems, Changes caused by agriculture and over grazing, effect of modern agriculture, fertilizer-pesticide problem, water logging, salinity, case studies.
 - f. Land resources: Land as a source, land degradation, man induces landslides, soil erosion and desertification.
 - g. Role of an individual in conservation and natural resources. Equitable use of resources for sustainable life style.
3. Eco system
 - a. Concept of an eco system. Structure of function of an eco system. Producer, consumer and decomposer. Energy flow in the eco system. Ecological succession. Food chain, food web and ecological pyramids. Introduction, Types, Characteristic feature, structure and function of following eco system. Forest ecosystem. Grass land eco system. Desert ecosystem. Aquatic ecosystem (ponds, stream, lakes, river ocean, estuaries)
4. Biodiversity and its conservation
 - a. Introduction - Definition, genetic, species and ecosystem diversity. Biographical classification in India. Value of biodiversity: consumptive use, productive use, social, ethical, esthetic and option value. Biodiversity at global, national and local levels.
 - b. India as a mega diversity nation. Hot-spot biodiversity. Threat to biodiversity: habit loss, poaching of wild life, man wild life conflicts.

- c. Endangered and endemic species of India. Conservation of biodiversity: In-situ and ex-situ conservation of biodiversity.

5. Environmental pollution

- a. Definition, Causes, effect and control measures of:-
 - i. Air pollution
 - ii. Water pollution
 - iii. Soil pollution
 - iv. Marine pollution
 - v. Noise pollution
 - vi. Thermal pollution
 - vii. Nuclear hazard
- b. Solid waste management: causes, effects and control measures of urban and industrial wastes.
- c. Role of an individual in prevention of pollution. Pollution case studies.
- d. Disaster management: floods, earthquakes, cyclone and landslides.

6. Social issues and the environment

- a. From unsustainable to sustainable development. Urban problems related to energy. Water conservation, rain water harvesting, watershed management.
- b. Re-settlement and rehabilitation of people its problems and concerns, case studies.
- c. Environment ethic, issues and possible solutions
- d. Climate change, global warming, acid rain, ozone layer, depletion, nuclear, accidents and holocaust, case studies. Wasteland reclamation.
- e. Environment protection Act: Air (prevention and control of pollution) Act, Water (prevention and control of pollution) Act, Wild life protection Act, Forest conservation Act.
- f. Issues involved in enforcement environment legislation. Public awareness.

7. Human population and the environment

- a. Population growth, variations among nations
- b. Population explosion, family welfare program. Environment and human health. Human rights. Value education.
- c. HIV AIDS. Women and child welfare. Roll of information technology in environment and human health. Case studies.

8. Field work

- a. Visit to local area to document environmental assets – river/forest/grass-land/hill/mountain
- b. Visit to local polluted site urban/rural/industrial/agricultural
- c. Study of common plants, insects, birds
- d. Study of simple ecosystems pond, river, hill slopes etc.

Books References

1. Environmental Biology – by KC Agarwal
2. The biodiversity of India
3. Hazardous waste – by Brunner RC
4. Marine pollution – by Cark RS
5. Environmental Encyclopedia – by Cunningham WP
6. Environmental Chemistry – by De AK
7. Water in crisis – by Gleiek HP
8. Encyclopedia of Indian Natural History – by Hawkins RE

SEMESTER – III

PAPER – I

PATHOLOGY, MICRO BIOLOGY AND PHARMACOLOGY

Instruction Hrs:

Theory – 100 hrs

Section – A: Pathology & Microbiology

1. Aims and objectives of study of pathology
2. Brief outline of cell injury, degeneration, necrosis and gangrene.
3. Inflammation: Definition, Vascular and cellular phenomenon difference between Transudate and exudates. Granulomas.
4. Circulatory disturbance: Hemorrhage, Embolism Thrombosis. Infarction, shock Volkmann's Ischemic contracture.
5. Neoplasia: Definition, characteristic features
 - a. Benign and malignant Tumor
 - b. Spread of tumors
6. General approach to immunity and hypersensitivity Reaction SLE.
7. General approach to Bacterial, Viral mycotic and parasitic infection with special importance to Tuberculosis. Syphilis and leprosy.
8. Diabetes Mellitus
9. Blood disorder: Anemia, Leukemia
 - a. Bleeding disorder
10. CVS: Heart and Blood vessels
 - a. Rheumatic heart disease, coronary heart diseases
 - b. Aneurysm Atherosclerosis
11. Respiratory System
 - a. COPD: Chronic Bronchitis, Emphysema, Asthma, Bronchiectasis
12. Bones and joint: Rheumatoid arthritis, Septic arthritis
 - a. Osteoarthritis, Spondyloarthropathy
 - b. Including ankylosing spondylitis
 - c. Bone tumour

13. Skin: Scleroderma, psoriasis
14. PNS and Muscles: Neuropathies
 - a. Poliomyelitis
 - b. Myopathies
15. CNS: Infection, Malformation, CVA
 - a. Demyelinating disease
 - b. Degenerative disease
 - c. CNS tumors

Section – B: Pharmacology

1. General Pharmacology
2. Autonomic Pharmacology
3. Drugs acting on CNS
4. Drugs acting on CVS
5. Drugs acting on respiratory system
6. Antibiotics & Chemotherapeutic agents
7. Hormones and drugs affecting endocrine function
8. Drugs acting on G.I system
9. Immuno modulators
10. Vitamins
11. Heavy metals & antagonists
12. Diagnostic agents

PAPER – II

BIOMECHANICS & KINESIOLOGY

Instruction Hrs:

Theory – 100 hrs

1. Essential concepts
 - a. Motion and forces
 - b. Force distribution - linear force, resultant force & equilibrium, parallel forces in one plane.
 - c. Newton's laws - Gravity and its effects on human body
 - d. Moments
 - e. Forces and moments in action
 - f. Concepts of static equilibrium and dynamic equilibrium
 - g. Composition & resolution forces
 - h. Friction
2. Kinematic concepts
3. Kinetic aspects of limb movement
 - a. Classification levers
 - b. Physiological significance of negative mechanical advantage
 - c. Muscle function and performance
4. Lower extremity kinematics
 - a. Hip and thigh
 - b. Hip joint motion and forces of hip joint
 - c. Two leg stances and one leg stances
 - d. Varus and valgus of femoral neck
 - e. Other factors affecting hip joint forces
 - f. Effect of cane by lever approach
5. Knee and leg kinematics
 - a. Motion of knee joint
 - b. Forces of knee joint
 - c. Patellofemoral joint
6. Ankle and foot kinematics
 - a. Motion of ankle
 - b. Forces of ankle joint
 - c. Stability of ankle joint
 - d. Weight bearing of foot
 - e. Arches of foot

7. Biomechanics of gait
 - a. Gait
 - b. Parameters of gait
 - c. Myokinetics of human gait
 - d. Gait deviation
 - e. Crutch and cane exercises
 - f. Biomechanics of walking aids
8. Posture
 - a. Anatomical aspects of posture
 - b. Factors affecting posture
9. Biomechanics of upper extremity
 - a. Shoulder joint
 - b. Elbow joint
 - c. Wrist joint and hand
 - d. Vicarious movement
10. Biomechanics of spine

PAPER – III
COMMUNITY MEDICINE

Instruction Hrs:
Theory – 50 hrs

1. Outline the objectives.
 - a. Outline the various measures of prevention and methods of intervention especially for diseases with disability.
 - b. Outline the national care delivery system and the health administration system of the central and state level.
 - c. Outline selected national health programs.
 - d. Define occupational health and list methods of prevention of occupational diseases and hazards.
 - e. Outline the employees' state Insurance scheme and its various benefits.
 - f. Describe the social security measures for protection from occupational hazards, accidents, diseases and the workman compensation act.
2. Family welfare program.
 - a. Define community based and institution - based rehabilitation. Describe the advantages and disadvantages of institution and community based rehabilitation.
 - b. Describe the following communicable diseases with reference to reservoir, mode of transmission, route of entry and levels of prevention.
 - i. Poliomyelitis
 - ii. Meningitis
 - iii. Encephalitis
 - iv. Tuberculosis
 - v. Filariasis
 - vi. Leprosy
 - vii. Tetanus
 - viii. Measles
 - c. Describe the Epidemiology of Rheumatic heart disease. Cancer, Chronic degenerative disease and cerebrovascular accidents.
 - d. Outline the influence of nutritional factors such as protein energy and malnutrition, anemia, vitamin deficiency and mineral disability.
 - e. List of principle of health education in rehabilitation services.
 - f. Define the role of community leaders and health professional in health education.
 - g. Outline the role of international health agencies in rehabilitation of the disabled.

SEMESTER – IV

PAPER – I

SURGERY – I

Instruction Hrs:

Theory – 100 hrs

Practical & Demonstration – 100 hrs

Section – A: General Surgery, Obstetrics & Gynecology

1. General Surgery

- a. Principles of general surgery and anaesthesia including blood transfusion and physiological response of the body to surgery.
- b. Principles of pre and postoperative management of surgical patients.
- c. Role of physiotherapy in general surgery.
- d. Describe the abdominal surgical incisions.
- e. Outline the post operative complications and management in:
 - i. Nephrectomy
 - ii. Appendectomy
 - iii. Herniorraphy
 - iv. Mastectomy
 - v. Thyroidectomy
 - vi. Colostomy
 - vii. Adrenalectomy
 - viii. Cystectomy
 - ix. Hysterectomy
 - x. Ileostomy
 - xi. Prostatectomy
 - xii. Cholecystectomy

2. Obs & Gyn

- a. Pregnancy – stage of pregnancy
- b. Labour – stages of labour – delivery
- c. Common gynecological problems.

Section – B: Cardiothoracic Surgery and Plastic Surgery

1. Cardiothoracic Surgery

- a. Incisions for cardiothoracic surgery
- b. General pre and post operative management of cardio - thoracic surgery
- c. Various surgical procedures for various and cardiac conditions/ diseases

2. Plastic Surgery

- a. Burn, Degrees of Burns
- b. Managements and Reconstructive Surgery following burns
- c. Complication of burns
- d. Types of skin Graft and flaps
- e. Principles of tendon transfer / transplant
- f. Cosmetic surgery
- g. Surgery of the hand with emphasis on reconstructive surgery and replantation surgery in trauma and leprosy

PAPER – II
MEDICINE – I

Instruction Hrs:
Theory – 150 hrs
Practical – 100 hrs

1. General Medicine
 - a. Definition
 - b. Etiopathogenesis
 - c. Pathology
 - d. Clinical features
 - e. Diagnosis
 - f. Differential Diagnosis
 - g. Principles of managements
2. Introduction to medicine
3. General principles of patient evaluation and management including over all idea about use of laboratory and imaging techniques.
4. Diseases of Respiratory system
 - a. Approach to a patient with Respiratory disease
 - b. Chronic obstructive Pulmonary Disease
 - c. Bronchial asthma
 - d. Pneumonia
 - e. Lung abscess
 - f. Bronchiectasis
 - g. Pleural effusion
 - h. Empyema Thoracis
 - i. Pneumothorax
 - j. Pulmonary tuberculosis
5. Diseases of GIS hepatobiliary Disorders
 - a. Approach to a patient of GIS Disease
 - b. U.G.IT. bleed
 - c. Jaundice
 - d. Viral Hepatitis
 - e. Cirrhosis of liver
6. Disease of Kidney
 - a. Approach to a patient of Renal disease
 - b. Acute glomerulonephritis
 - c. ARF, CRF
 - d. Nephrotic Syndrome

7. Hematologic disease
 - a. Approach to a patient of hematologic disease
 - b. Anaemia: Iron deficiency anaemia, Haemolytic anaemia, Megaloblastic anaemia
8. Oncology
 - a. Lymphomas
9. Endocrine and metabolic disease
 - a. Acromegally, gigantism & dwarfism
 - b. Hypothyroidism & hyperthyroidism
 - c. Adrenal hyper function & hypofunction
 - d. Diabetes mellitus
 - e. Hypoglycaemia
 - f. Vitamine D & calcium metabolism
 - g. Parathyroid gland disorder
10. Nutritional disease
 - a. Obesity
 - b. Protein energy malnutrition
11. Connective tissue disease
 - a. Approach to patient to connective tissue disease
 - b. Rheumatoid arthritis
 - c. Gout
12. Infectious disease
 - a. Malaria
 - b. Filaria
 - c. Tetanus
 - d. Fungal infection
13. HIV & AIDS
14. Diseases of Skin
 - a. Scabies
 - b. Fungal infection
15. Disease due to environmental factor & poisons
 - a. Heat stroke
 - b. Radiation injury
 - c. Snake bite
 - d. Organophosphorus poisoning
 - e. Oleander poisoning

PAPER – III

MEDICINE – II (CARDIOLOGY & PAEDIATRICS)

Instruction Hrs:
Theory – 150 hrs

Section – A: Cardiology & Work physiology (100 hrs)

1. Cardiology
 - a. Basic anatomy of heart, coronary circulation and development of heart.
 - b. Normal cardiac contraction and relaxation: mechanism and diagnosis.
 - c. Acute rheumatic fever: etiology, diagnosis and management.
 - d. Valvular heart disease: diagnosis and management.
 - i. Mitral stenosis
 - ii. Mitral regurgitation
 - iii. aortic regurgitation
 - iv. Aortic stenosis
 - e. Ischemic heart disease: Clinical features, diagnosis and management.
 - f. Hypertension: classification and treatment.
 - g. Congestive heart failure: etiology diagnosis and management.
 - h. Peripheral vascular disease, deep vein thrombosis: etiology and management.
2. Work Physiology
 - a. Physiology of exercises
 - b. Cardiac output and cardiac cycle during regulation
 - c. Cardiac rate during exercise
 - d. Oxygen consumption of the body at rest, during exercise and after exercise
 - e. Effect of exercise on:
 - i. Calorie intake
 - ii. Coronary circulation
 - iii. Metabolism
 - iv. Renal blood flow
 - v. Contractility of myocardium
 - vi. Blood pressure
 - vii. Haemo dynamics variable
 - viii. Increase in carbon dioxide tension and mixing venous blood
 - ix. Increase in pulmonary ventilation
 - f. Equipment for work physiology
 - i. Ergo meter – cycle type
 - ii. Ergo meter – treadmill type

Section – B: Pediatrics (50 hrs)

1. Describe growth and development of a child from birth to 12 year including physical, social, adaptive development.
2. List the maternal and neonatal factors contributing to high risk pregnancy.
3. The neonate: inherited diseases, maternal infection viral and bacterial, maternal diseases incidental to pregnancy induced hypertension, chronic maternal diseases such as heart diseases, renal failure, tuberculosis, diabetes, epilepsy, bleeding to mother at any trimester.
4. Briefly describe community programs; international (WHO), national and local for prevention of poliomyelitis, blindness, deafness, mental retardation and hypothyroidism. Outline immunization schedule for children.
5. Cerebral palsy: Define and briefly outline etiology of prenatal, perinatal, postnatal causes. Briefly mention pathogenesis, types of cerebral palsy (classification), findings on examination, general examination of C.N.S, musculoskeletal and respiratory system.
6. Briefly outline associated defects: mental retardation, microcephally, blindness, hearing and speech impairment, squint and convulsions.
7. Prevention: appropriate management of high risk pregnancies, prevention of neonatal and postnatal infections, metabolic problems.
8. Muscular dystrophy: outline various forms, modes of inheritance and clinical manifestation physical finding in relation to disabilities progression of various form and prognosis. Describe treatment goals in forms which are and are not fatal.
9. Spina bifida, meningomyelocele: Outline development, clinical features - lower limbs, bladder and bowel control, complications UTI & hydrocephalus, medical treatment and surgical treatment.
10. Still's disease: classifications, pathology in brief, physical finding, course & prognosis. Outline treatment, prevention and correction of deformity.
11. Acute C.N.S infections: classify (Bacterial and viral) and outline the acute illness. CSN sequelae leading to mental retardation, blindness deafness, speech defect, motor paralysis, bladder and bowel problems, seizure disorder and specific problems such as subdural effusion hydrocephalus, pres- sure sores feeding difficulties.
12. Normal diet of new born and child list dietary calorie, fat protein, mineral and vitamin requirement in a normal child and in a child with malnutrition. Classify and outline etiology, findings and treatment of rickets vitamin D deficiency and resistant rickets.

PAPER – IV

PHYSIOTHERAPY IN CARDIO-PULMONARY CONDITIONS

Instruction Hrs:

Theory – 200 hrs

Practical & Demonstration – 100 hrs

Section – A: Physiotherapy in Cardiovascular Conditions (150 hrs)

Theory (100 hrs)

Practical (50 hrs)

1. Cardiac anatomy
2. Cardiac physiology
3. Congenital heart diseases
4. Acquired heart diseases
5. Ischemic heart diseases
6. Life span development cardiovascular system
7. Peripheral vascular diseases, Physiotherapy management for common arterial venous and lymphatic conditions
8. Evaluation for cardiovascular system
9. Cardio pulmonary resuscitation
10. Cardiac rehabilitation
11. Incisions for cardiac surgeries, drainage tubes and bottles, ventilators – uses and functions of ventilators
12. Post-operative complications, Pre and Post operative physiotherapy management of open heart and close heart surgeries.

Section – B: Physiotherapy in Pulmonary Conditions (150 hrs)

Theory (100 hrs)

Practical (50 hrs)

1. Respiratory anatomy and physiology.
2. Respiratory pathology of obstructive restrictive and infective conditions.
3. A comparative neonatal and pediatric respiratory anatomy, physiology with adults.
4. Respiratory assessment
5. Respiratory investigations
6. Breathing strategies, chest clearance techniques, exercises testing and training. Managing chest conditions using these conditions.
7. Incisions for pulmonary surgery: Drainage tubes and bottles, ventilators - uses and functions of ventilators.

8. Post-operative respiratory complication, Physiotherapy for pulmonary surgeries, Pre and Post operative physiotherapy management of the following condition:
 - a. Thoracotomy
 - b. Lobectomy
 - c. Thoracoplasty
 - d. Pneumonectomy
9. Management of atelectasis, pneumothorax, fistula
10. Monitoring in ICU
11. Role of physiotherapy in ICU
12. Pulmonary rehabilitation
13. Pediatric chest physiotherapy: Outline the clinical findings, complications and medical treatment of bronchiectasis, lung abscess and bronchial asthma, cystic fibrosis, primary complex in infant and children. Acute pediatric respiratory distress syndrome intensive pediatric care. Intensive neonatological and pediatric surgical care Cardio respiratory rehabilitation in children.

SEMESTER – V

PAPER – I

ALTERNATIVE MEDICINE

Instruction Hrs:

Theory – 150 hrs

Practical & Demonstration – 50 hrs

1. Yoga: Definition, History, Principles, Concepts, General effects of yoga posture on musculo-skeletal system. Specific effects of individual Yogic posture on musculo-skeletal system. Yoga and therapy rationale.
2. Naturopathy: Definition, History, Principles, Concepts, General effects of Naturopathy.
3. Acupuncture & Acupressure:
 - a. Acupuncture points and meridians - their function extra meridians
 - b. Forbidden points
 - c. Complication
 - d. Contraindications
 - e. Specific important points (Luo and source points horary points)
 - f. Mechanism of acupuncture - physiology
 - g. Techniques
 - h. Equipment
 - i. Methods for asepsis, sterilization
 - j. Relevance of acupuncture to physical therapeutic Trigger points
 - k. Electro acupuncture

PAPER – II

PHYSIOTHERAPY IN SURGICAL CONDITIONS

Instruction Hrs:

Theory – 200 hrs

Practical – 100 hrs

1. Abdominal Surgery: Pre and post operative Physiotherapy management of the following abdominal surgical conditions (Incision, pre and post operative complications must be explained).
 - a. Total gastrectomy
 - b. Partial gastrectomy
 - c. Appendectomy
 - d. Herniorraphy
 - e. Cholecystectomy
 - f. Hysterectomy
 - g. Radical mastectomy
 - h. Colostomy
2. Obstetrics and Gynaecology:
 - a. Antenatal and post natal training
 - b. Prolapsed uterus
 - c. Urogenital dysfunction
 - d. Pre and post operative management of pelvic floor surgery
 - e. Common gynecological conditions
3. Plastic Surgery and Burns:
 - a. Degrees of burns
 - b. Physiotherapy management of burns
 - c. Pre and post operative physiotherapy of skin grafting, reconstructive surgery of hand, Tendon transfer etc.

SEMESTER – VI

PAPER – I

MEDICINE – III (NEUROSCIENCE)

Instruction Hrs:

Theory – 175 hrs

Practical & Demonstration – 100 hrs

1. General principles of neurological diagnosis
2. Cerebro vascular diseases: Cerebral vascular accident, Cerebral thrombosis, embolism & hemorrhage
3. Intra cranial tumors
4. Acute infection of CNS: Encephalitis, Meningitis, poliomyelitis
5. Traumatic injury of the Head & spine
6. Parkinsonism and other extrapyramidal disorders
7. MS & other demyelinating diseases
8. ALS (amyotrophic lateral sclerosis) and other motor neuron diseases
9. Diseases of Peripheral Nerves, cranial nerves, G.B.S including Peripheral nerve injury & principles of management
10. Myasthenia Gravis
11. Diseases of muscles
12. Seizures and epilepsy
13. Headache
14. Dementia
15. Cerebral Palsy
16. Principle of management of cranial & spinal trauma
17. Development anomalies of CNS & their brief management
18. Degenerative diseases of spine & outline of management
19. Management of pain syndromes
20. Outline of clinical presentation & management of brain tumors & spinal cord compressions.
21. Neurosurgical intensive care study
22. Use of operative microscope, endoscopy, stereotactic surgery, minimally invasive surgery in Neurosurgical perspective.
23. Rehabilitation of neurologically disabled patients

PAPER – II

MEDICINE – IV (PSYCHIATRY)

Instruction Hrs:
Theory – 100 hrs

1. Neurosciences, Neuroanatomy, Neurotransmitter study etc.
2. Examination and diagnosis of psychiatric cases.
3. Clinical manifestations of psychiatric disorders.
4. Classification of mental disorders.
5. Theories of personality and psychoanalysis.
6. Neuropsychiatric aspects of
 - a. Cerebrovascular disorders
 - b. Brain tumors
 - c. Epilepsy
 - d. Traumatic brain injury
 - e. Movement disorders
 - f. Multiple sclerosis
 - g. HIV infection and AIDS
 - h. Headache
 - i. Neuromuscular disorders
7. Delirium, Dementia, Amnesic and cognitive disorders.
8. Substance related disorders alcohol, amphetamine, cannabis, opioid, caffeine, nicotine, hallucinogens etc.
9. Schizophrenia
10. Other psychotic disorders
 - a. Schizo-affective disorders
 - b. Schizophreniform and brief psychotic disorders
 - c. Delusional disorders, shared psychotic disorder
 - d. Acute and chronic psychotic disorders
 - e. Postpartum psychotic syndromes
11. Mood disorders
12. Anxiety disorders - GAD, phobias, panic disorders, ASD, PTSD, OCP
13. Somatoform disorders
 - a. Conversion disorders
 - b. Somatization disorders
 - c. Hypochondriasis
 - d. Pain disorders
 - e. Body dysmorphic disorders
 - f. Chronic fatigue syndrome
14. Factitious disorder

15. Dissociative disorder
16. Normal human sexuality and gender identity disorders
17. Eating disorders
18. Sleep disorders
19. Impulse control disorders not classified elsewhere
20. Adjustment disorders
21. Personality disorders
22. Psychological factors affecting medical conditions
23. Disaster-types, psychiatric co-morbidities and management
24. Biological therapies
 - a. Dopamine receptor antagonists
 - b. Serotonin-dopamine antagonist
 - c. Benzodiazepine receptor agonists 2 and antagonists
 - d. Mood stabilizers-lithium, valproate, carbamazepine etc.
 - e. Tricyclics and tetracyclics
 - f. Selective serotonin reuptake inhibitors
 - g. SNRI
 - h. Antihistaminic
 - i. Electro convulsive therapy
25. Mental retardation
26. Suicide
27. Early onset schizophrenia
28. Attention deficit disorder
29. Conduct disorders
30. Tic disorders
31. Feeding and eating disorders of infancy and early childhood
32. Psychotherapies
33. Medical ethics

PAPER – III

SURGERY – II (ORTHOPAEDICS)

Instruction Hrs:

Theory – 100 hrs

Practical & Demonstration – 50 hrs

1. Fractures and dislocation including soft tissue injuries
 - a. Pathology of fractures and repairs of bones
 - b. Reasons for union, non-union, delayed union, fibrous union and myositis
 - c. Common fractures of upper extremity, lower extremity including spine – Management, Complications etc.
 - d. Dislocation of shoulder, elbow, hip, knee, spine
 - e. Rupture, contusion and sprain of muscles, tendons and ligaments.
 - f. Knee injuries – Injury to medial ligament, internal derangement and meniscus tear
 - g. Ankle injuries – Lateral ligament sprain of ankle
2. Deformities: common congenital and acquired deformities of spine, foot, knee, hip, shoulder, elbow, wrist and hand. Cervical rib, torticollis, metatarsalgia, claw hand.
3. Inflammatory conditions and lesions of joints and bones. Osteomyelitis, tuberculosis, pyogenic infection, osteoarthritis, RA, TB Joints, tenosynovitis, synovitis, capsulitis, tendonitis, osteoporosis, osteomalacia, sciatica, low back pain, brachial neuralgia.
4. Operative procedures
5. Management after A.O. fixation
6. Bone tumor: classification and management
7. Management of open wound with external fixator

SEMESTER – VII

PAPER – I

REHABILITATION SCIENCE

Instruction Hrs:

Theory – 150 hrs

1. Introduction
 - a. Define the term rehabilitation. Explain its aims and principles, scope of rehabilitation.
 - b. Discuss team work involved in rehabilitation, explaining briefly the role of each team member.
2. Therapeutic techniques
 - a. Agencies involved in rehabilitation of a physically handicapped
 - b. Legislation for physically handicapped (in brief)
 - c. Limitations of each team member in rehabilitation of a physically disabled individual.
3. Communication problems
 - a. Identify communication problems, classify these and outline principles of treatment, outline of speech therapy and hearing aids.
4. Behavioral problems
 - a. Identify behavioral problems in the disabled and outline the principles of management.
5. Mobility aids
 - a. Demonstrate knowledge of the indications for different types of mobility aids and their functions, e.g. wheelchairs, walkers, crutches.
6. Pre-vocational evaluation
 - a. Discuss methods and team involvement in pre - vocational evaluation and training.
7. Architectural barriers
 - a. Describe architectural barriers and possible modifications with reference to rheumatoid arthritis, cerebrovascular accident, spinal cord injury and other disabling conditions.
8. Disability evaluation
 - a. Outline the principles of disability evaluation and discuss its use.

9. Legal aspects

- a. Outline the principles of disability in terms of compensation for disability and benefits available to the disabled.

10. Social implications

- a. Outline the social implications of disability for the individual and for the community.

11. Community based rehabilitation module

- a. Describe a DBR MODULE and compare this with an institutions based rehabilitation system.

12. Visual disability

- a. Definition and classification, mobility technique communication skills, sensory re-education, emotional and psychological aspects of blindness, facilities for blind, prevention of blindness.

13. Mental retardation

- a. Definition and classification, prevention and existing facilities for mentally retarded children.

14. Outline of social and vocational counseling.

15. Classification of aids and appliances

- a. Measurement of P.O.P. Cast techniques.
- b. Simple splints techniques.

16. Principles and check out procedures for static, dynamic alignment and training

- a. Spinal orthosis
- b. L.L Orthotics & prosthetics
- c. U.L Orthotics & prosthetics

PAPER – II

PHYSICAL DIAGNOSIS & PHYSICAL FITNESS

Instruction Hrs:
Theory – 200 hrs
Practical – 100 hrs

Section – A: Physical diagnosis

1. Problem oriented medical record – history, concept, advantages
2. Communication with the patient – Principal and methods
3. Physical Diagnosis on the basis of:
 - a. Musculoskeletal system
 - i. Maitland's concept
 - ii. Kaletenborn concept
 - iii. Cyriax approach
 - iv. Mckenzie's concept
 - v. Neural tension tests – normal & abnormal findings
 - b. Neurological system (for CNS problem)
 - i. Motor learning
 - ii. Bobath approach (normal movement concept)
 - iii. Vojita approach
 - c. Clinical reasoning and clinical decision making
 - d. Rational of plan of treatment

Section – B: Physical fitness & Ergo therapeutics

1. Factors responsible for occupational hazards-stress, faulty working conditions (Biomechanical aspects) – Thermal stress, over-use, pollution, noise, air, water, food.
2. Accidents-electrical, mechanical, thermal, chemical
3. Disability evaluation (functional) interpretation and legislation principles, techniques, suggestions for compensation
4. Ergonomic evaluation-evaluation of working area, type of work, fitness testing for the same.
5. Preventive P.T measures
6. Fitness programs for specific work
7. Sports and industry
8. Planning, developing and management towards work efficiency, productivity, avoidance of accidents and other use.
9. Relaxation program for stress

PAPER – III

PROFESSIONAL MANAGEMENT & ETHICS

Instruction Hrs:
Theory – 100 hrs

Section – A: Professional ethics & legal issues

1. The implications and confirmation to the rules of professional conduct.
2. Legal responsibility for their actions in the professional context understanding liability and obligation in case of medico - legal action.
3. A wider knowledge of ethics relating to current social and medical policy in the provision of health care.
4. National and international professional association, and education body - difference between scientific association (Professional body) and statutory body.
5. The role of international health agencies such as WHO.

Section – B: Management studies

1. Definition – Branches of management, principles of health sector management.
2. General principles Management – Theories of management, Basic concept and theories.
3. Personnel management – Policies and procedures, Basic concepts and theories.
4. Financial issues including budget and income generation.
5. Principles of an organization chart.
6. Organization of department – Planning, space, manpower, materials, basic requirements.
7. Resource and quality management – Planning with change and coping with change.
8. Self - management:
 - a. Preparing for first job
 - b. Time Management
 - c. Career development

PAPER – IV

**RESEARCH METHODOLOGY, BIOSTATISTICS AND INTRODUCTION TO
COMPUTER SCIENCE**

Non-University examination

Instruction Hrs:

Theory – 50 hrs

1. Review of literature
2. Study design
3. Sample size
4. Sampling variability & significance
5. Protocol writing
6. Ethical aspects
7. Data collection analysis, interpretation and presentation
8. Common statistical terms
9. Measures of location, average & percentiles
10. Variability & its measures
11. Normal distribution & normal curve
12. Probability
13. Significance of difference in mean
14. Chi - square test
15. Correlation & regression
16. Demography & vital statistics
17. Correlation of measures of population & vital statistics
18. Use of micro computer in research

SEMESTER – VIII

PAPER – I

PHYSIOTHERAPY IN NEUROLOGICAL CONDITIONS

Instruction Hrs:

Theory – 100 hrs

Practical – 150 hrs

1. Introduction
 - a. Brief review of the following medical conditions and various modalities of physiotherapy, aims, means and techniques physiotherapy should be taught.
2. UMN lesions
 - a. Hemiplegia
 - b. Cerebral palsy
 - c. Multiple sclerosis
 - d. Monoplegia, Paraplegia, Tetraplegic Syndrome
 - e. Sub - acute combined degeneration of spinal cord
 - f. Syringomyelia
 - g. Transverse myelitis
 - h. Parkinsons's disease
 - i. Extra pyramidal lesions
 - j. Motor neuron disease
 - k. Ataxia
 - l. Tabes dorsalis
 - m. Acute CNS infection
 - n. Pre and post operative management and complications of Head injury
Laminectomy, surgery following brain tumor etc.
3. Muscle disorders and LMN lesions
 - a. Myopathy and muscular dystrophies
 - b. Poliomyelitis
 - c. Polyneuritis
 - d. Peripheral neuropathy, Brachial neuritis and neuralgia
 - e. Facial palsy and bell's palsy
 - f. Leprosy
 - g. Peripheral nerve injuries, Erb 's palsy
 - h. Sciatica
 - i. Peripheral nerve injuries - (Non-operative) - pre and post operative management of nerve repair and grafting.
 - j. General and physiotherapeutic management of psychiatric patients.

PAPER – II

PHYSIOTHERAPY IN MUSCULOSKELETAL CONDITIONS

Instruction Hrs:

Theory – 100 hrs

Practical – 150 hrs

1. Introduction
 - a. Brief review of the following surgical conditions and various physiotherapy modalities, aims, means and techniques of physiotherapy should be taught.
2. Traumatology: general physiotherapeutic approach for traumatic conditions, Fractures and Dislocation.
 - a. Classification, types of displacement, methods of immobilization.
 - b. Common sites of fracture. Healing of fractures and factors influencing union, non – union, delayed union etc.
3. Specific fractures and their complete physiotherapy management
 - a. Upper limb – clavicle, humerus, radius and ulna. Crush injuries of hand.
 - b. Lower limb – Fracture neck of femur, shaft of femur, patella, tibia and fibula, pott's fracture of tarsal and metatarsal bones.
 - c. Management of fracture spine with (paraplegia) as well as without neurological deficit.
4. Dislocation of shoulder, hip, ACJ, SCJ, elbow.
5. Soft tissue injuries, synovitis, capsulitis, Volkmann's ischemic contracture etc.
6. Tear of semilunar cartilage and cruciate ligaments of knee. Sprain, strain, overuse syndrome, tendinitis, tendinosis, bursitis.
7. Amputations: levels of amputation of upper and lower extremity - stump care, stump bandaging, pre and post fitting prosthesis management (check out of prosthesis, training etc.)
8. Deformities
 - a. Congenital: torticollis and cervical rib, C.T.E.V., pes cavus, pes planus and other common deformities.
 - b. Acquired: scoliosis, kyphosis, lordosis, coxa vara, genu valgum, genu varum and genu recurvatum.
9. Degenerative and infective conditions: Osteoarthritis of major joints, spondylosis, spondylitis, spondylolisthesis prolapsed intervertebral disc lesion. Periarthritis (rotator cuff lesion) of shoulder, tuberculosis of spine, bone and major joints, perthes disease.

10. Rheumatoid arthritis, ankylosing spondylitis, psoriatic arthritis, syphilitic arthritis, scleroderma etc. and other miscellaneous orthopedic conditions commonly treated by physiotherapy.
11. Growth and development, maternal and neonatal factors contributing to high risk baby, cp, myopathy, spina bifida, hydrocephalus, still's disease. CTEV, CDH, Arthro Gyphosis Multiplex Congenita (AGMC), rickets, torticollis, osteogenesis imperfecta.

PAPER – III

PROJECT BASED ON RESEARCH METHODOLOGY AND BIOSTATISTICS

Instruction Hrs:

Theory – 150 hrs